

[REDACTED]

LIMITED-SOURCES JUSTIFICATION (LSJ)
Animatronic TCCC manikins

1. Defense Logistics Agency (DLA) Troop Support, Medical Supply Chain, Hospital Supply Division, is the contracting activity.
2. The justification is for acquisitions supporting the United States Navy (USN) in Coronado, California; The customer is seeking to acquire Animatronic Tactical Combat Casualty Care (TCCC) manikins: TACMED SOLUTIONS: **PATIENT SIMULATOR**. The order can be placed against ECAT Solicitation SPE2DH-21-R-0002 Contract Number SPE2DH-24-D-0007, which is valid through March 14, 2029.
3. The acquisitions are for the purchase of the following:

Manufacturer: TACMED SOLUTIONS

Item Description: PATIENT SIMULATOR - CLINICAL RESPONSE UPPER RESUSCITATE (CRU-R)

Manufacturer Part Number: FG-CRU-R-0002

Quantity: 8 Each

Unit Price: [REDACTED]

Manufacturer: TACMED SOLUTIONS

Item Description: PATIENT SIMULATOR - HEMO - LOWER BODY

Manufacturer Part Number: FG-HEMO-0004

Quantity: 8 Each

Unit Price: [REDACTED]

Manufacturer: TACMED SOLUTIONS

Item Description: Accessories

Manufacturer Part Number: Multiple

Price: [REDACTED]

Total Estimated Dollar Value: [REDACTED]

4. The authority waiving the competition requirements is FAR 8.405-6(a)(1)(i)(B), only one source is capable of providing the supplies or services required at the level of quality required because the supplies or services ordered are unique or highly specialized. The use of this authority is necessary to meet the locations' requirements for 8 complete animatronic TCCC manikins (combined number between 8 upper resuscitate and 8 lower body). The manikins are necessary for providing high-quality advanced medical training across the entirety of the Department of War's (DoW) TCCC courses of instruction. These manikins allow locations to comply with DoW and Office of the Chief of Naval Operations (OPNAV)



[REDACTED]

training requirements and eliminate the need for contract training. Recently, DoW made TCCC mandatory across the military. Currently, locations do not have access to training aids that meet the requirements of the Council of TCCC. The Tacmed Solutions manikins offer features that are not found in other manikins.

The required animatronic TCCC manikins must satisfy the following minimum requirements established by the aforementioned location:

- a. **Advanced Cardiovascular Life/Basic Life Support Simulation** - The manikin's chest must measure resistance and recoil during CPR compressions. It must also feature palpable carotid and femoral pulse points, generated by software-controlled solenoid actuators that synchronize with a vital signs monitor.
 - b. **Airway Management Simulation** - The manikin's airway must be modeled on human anatomy, including structures such as teeth, tongue, and vocal cords. A pressure sensor must be embedded in the upper teeth. The airway must be equipped with a remote-controlled part that physically moves to create a partial or total obstruction.
 - c. **Cricothyroidotomy Simulation** - The manikin's neck must have palpable cartilage landmarks and a replaceable, multi-layered tissue cartridge at the cricothyroid membrane. A contact sensor must be positioned behind the membrane.
 - d. **Chest Tube Training** - The manikin's ribcage must have bilateral sites covered by replaceable, multi-layered tissue pads for incision and an internal proximity sensor.
 - e. **Needle Decompression** - The manikin must be equipped with multiple self-sealing tissue plugs located on the upper part of the chest (slightly below the collarbone) and on the side of the chest (below the armpit) on both sides of the torso. Each site must contain a pressurized air bladder with a pressure-release valve.
 - f. **Intraosseous Access Simulation Technology** - The manikin must have hardened inserts at the sternum and bilateral tibias, covered by skin-analogue pads. Each site must contain an embedded contact sensor.
 - g. **Penetrating Chest Wound Simulation Technology** - The manikin's torso must feature a pre-formed wound cavity and have a pressure sensor located in the wound.
 - h. **Inguinal Wound Simulation Technology** - The manikin must have a wound cavity in the inguinal region with a pressure sensor.
 - i. **Intubation with Error Detection** - The manikin must have a dual-passage airway. The trachea and esophagus must each have a sensor.
 - j. **Gunshot Wound with Bleeding Control** - The manikin must have wounds connected to an onboard fluid reservoir, with bleeding driven by a software-controlled pump. The software must be able to alter the fluid flow rate.
- [REDACTED]

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- k. **Amputation with Tourniquet Feedback Technology** - The manikin must feature a bleeding partial amputation site with an integrated tourniquet sensor.
 - l. **Leg Splinting and Immobilization Simulation Technology** - The manikin must have articulated limbs with internal gyroscopic sensors.
 - m. **Animatronic Patient Movement** - The manikin must contain high-torque servo motors integrated into the limb joints.
 - n. **Environmental and Durability Standards** - The manikin must have a water-resistant outer body, and internal chassis.

The animatronic TCCC manikins will be utilized by the US Navy forces located in Coronado, California.

The manikin has a chest that provides measurable resistance (depth) and recoil (compression rate and depth) during CPR compressions. An internal accelerometer and pressure sensor measure compression rate and depth. Palpable carotid and femoral pulse points are generated by software-controlled solenoid actuators. It possesses an airway modeled on human anatomy, which can be intubated, and its breathing passage can be intentionally blocked, preventing airflow (simulate airway obstruction). A teeth embedded pressure sensor reports force, and a triggered mechanical actuator creates obstruction. A sensor at the trachea confirms endotracheal tube depth, while a separate esophageal sensor detects and reports tube misplacement to the instructor's remote control. For cricothyroidotomy, it features a neck with a replaceable tissue cartridge at the cricothyroid membrane containing a contact sensor for airway entry confirmation. For chest tube training, the manikin's torso has bilateral chest sites with replaceable tissue pads and an internal proximity sensor; tube insertion in the pleural space is detected and logged as an event. The manikin can simulate needle decompression; this is an emergency procedure where a needle is inserted into the chest to release trapped air and relieve pressure around the lungs. The manikin's needle decompression sites contain pressurized air bladders with pressure-release valves that provide an audible hiss upon puncture, while the software registers the event. For intraosseous catheter placement leg simulation, the sternum and bilateral tibias have hardened inserts embedded with contact sensors; each placement is registered as an active line in the software. The manikin has penetrating wounds on different parts of the body - chest wound and an inguinal wound. Internal pressure sensors detect wound airtight seal placement (signaled to the software); the software signals the pump to stop bleeding only when sustained and adequate packing pressure is achieved. The manikin has a dual-passage airway; the trachea and esophagus each have a sensor where tube misplacement is detected and reported to the software. The manikin can simulate bleeding gunshot wounds using a software-driven pump and sensors; the fluid/blood flow rate automatically adjusts real-time for controlled/successful wound packing. The manikin can simulate tourniquet application for leg amputation; the bleeding partial amputation sites have an integrated sensor that measures and stops fluid/blood flow. For limb immobilization simulation, the articulated limbs contain gyroscopic sensors; these sensors register when a splint is attached and capture unsuccessful limb restraint to the software. The manikin has animatronic movements because of its high-torque servo (precise control) motors integration; the limb joints are a direct-output system giving instructors pre-programmed movement triggers including kicking and seizures. To ensure operational longevity/be water-resistant, the manikin's outer body is made of silicone while the internal chassis' s internal electronics are conformal coated or potted in epoxy.





The Tacmed Solutions animatronic TCCC manikin identified above is the only product currently available on ECAT or in the commercial market that meets all minimum clinical/technical requirements indicated above.

5. The Contracting Officer has determined the contract prices of supplies under the Equipment ECAT multiple award program fair and reasonable. [REDACTED] The Contracting Officer has determined that this acquisition represents the best value to the Government consistent with FAR 8.404(d).

6. The USN site team performed clinical requirements assessments through market research [REDACTED]

- [REDACTED]
- [REDACTED]
- [REDACTED]
- [REDACTED]
- [REDACTED]

[REDACTED]

Additional market research was completed by DLA Troop Support in January 2026 [REDACTED]

[REDACTED]

The TACMED SOLUTIONS: **PATIENT SIMULATOR** is available on ECAT via SPE2DH-24-D-0007.

7. ECAT vendors have the opportunity to add new products that fall within the scope of the Medical Equipment ECAT program on a quarterly basis. Therefore, any ECAT vendor may add products, including animatronic TCCC manikins, enhancing competition for future acquisitions.
8. DLA Troop Support continually performs market research of animatronic TCCC manikins to identify possible other acceptable sources for future acquisitions, in order to overcome the barriers that led to restricted consideration. Furthermore, DLA will continue monitoring the market, however, any new item considered for procurement would be required to meet the Government's minimum clinical requirements. In the event another vendor outside of the Equipment ECAT Program offers animatronic TCCC manikins, DLA will encourage any such vendor/manufacture to participate in the Equipment ECAT Program.
9. As the Contracting Officer, I hereby certify that the information contained in this Limited-Sources Justification is accurate and complete to the best of my knowledge and belief.

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EVAN LESSIN
Contracting Officer, FSHC

I hereby certify that the requirements that form the basis of this justification are accurate, that these supplies meet the Government's minimum needs, and that this justification is accurate and complete to the best of my knowledge and belief.

[REDACTED]

[REDACTED]

Product Specialist

[REDACTED]